



**GDAŃSK UNIVERSITY
OF TECHNOLOGY**

Trustworthy and Privacy-Preserving Protocols in Distributed Systems for Sensitive Services

Stanisław Barański

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- The AI wave raises the stakes: systems trained on sensitive data should be auditable, fair, and privacy-preserving by design.
- Privacy is not only a preference: it supports progress and social stability by reducing surveillance and coercion, while enabling safe data use (e.g., healthcare, AI).

This dissertation addresses how to build trustworthy and privacy-preserving systems for sensitive services using distributed trust and cryptographic guarantees.

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- Distributed Ledger Technologies (DLT) enable public coordination and audit without a single operator.
 - DLT is a replicated, append-only public log/state maintained by distributed consensus. It gives us a shared public transcript (T_{pub}): a globally visible sequence of protocol events (e.g., commitments, ciphertexts, proofs, signatures, receipts, and ledger state updates) that all parties can read and verify.

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- Combined: abuse becomes computationally infeasible, or requires corrupting many independent parties (often economically infeasible).

Sensitive services

- **Sensitive services** process protected data/relations (legal, regulatory, business, or personal sensitivity).
- In this dissertation we focus on the following set of five sensitive service classes:

$$\mathcal{S}_{sens} = \{s_{ivot}, s_{ppml}, s_{iot}, s_{anonser}, s_{mixer}\}.$$

- Each service instance s is identified by:
(f_s , T_{pub} , protected secrets/links, $\mathcal{R}_s$, threat model).
 - f_s : the service functionality (e.g., tally votes, aggregate data, mix funds).
 - T_{pub} : the public transcript (commitments, proofs, receipts).
 - Protected secrets/links: private inputs or relationships that must not be revealed.
 - $\mathcal{R}_s$: the set of roles/parties involved (voters, servers, clients, etc.).
 - Threat model: assumptions about adversarial capabilities and corruption thresholds.
- We model a service as a stateful functionality with a public transcript:

$$(y_1, \dots, y_n, T'_{pub}) \leftarrow f_s(x_1, \dots, x_n, T_{pub}).$$

Security properties

- Each sensitive service requires a specific set of security properties. We focus on three core properties:
 - **Correctness (CORR)**: outputs correspond to the intended logic and public transcript.
 - **Privacy (PRIV)**: protected inputs/links are not revealed beyond what is implied by public transcript and legitimate outputs.
 - **Liveness (LIVE)**: requests eventually produce an output or a terminal abort.

Role of VPPC and DLT

- VPPC enforces that outputs (y_i) correspond to the *committed private inputs* under f_s , without revealing inputs (x_i) or protected links.
- DLT enforces a single, consensus-ordered and tamper-evident public transcript T_{pub} of submitted inputs and outputs/proofs, so everyone agrees what was provided and can audit that only valid, verified state transitions were accepted.
- The main challenge is that:
 - DLT can enforce **Correctness** and **Liveness** for what is **publicly** visible and checkable on T_{pub} ,
 - but sensitive services need **Privacy** and **Correctness** for **private** inputs, hence the need for VPPC.

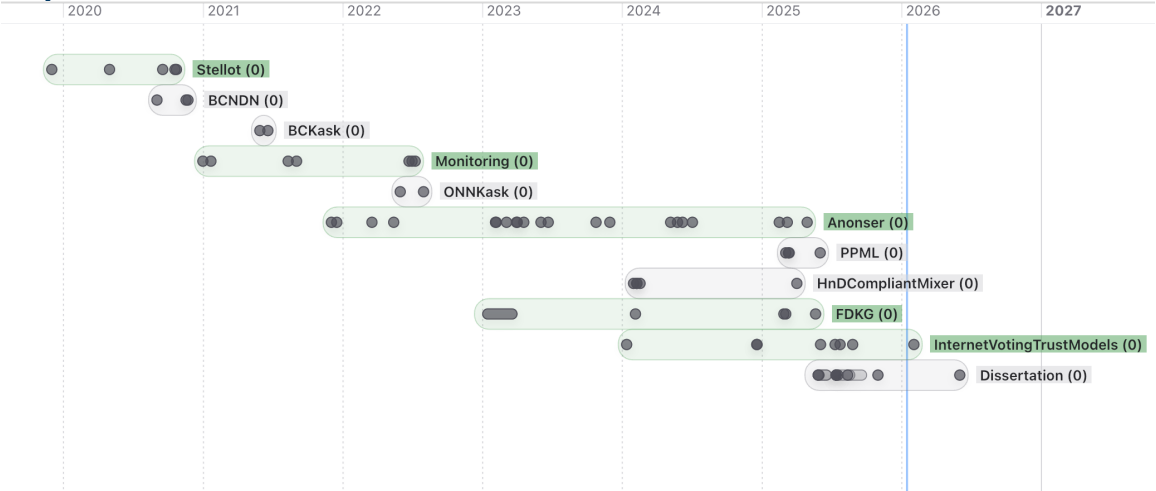
Protocols for sensitive services using VPPC and DLT

- $\mathcal{P}_{\text{VPPC+DLT}}$ is the set of protocols to realise sensitive services that combine:
 - a DLT layer to coordinate and provide the immutable public transcript (T_{pub}), and
 - a VPPC layer to do sensitive computation off-chain (or privately) and output proofs/attestations that the DLT can verify.

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 - a VPPC layer to do sensitive computation off-chain (or privately) and output proofs/attestations that the DLT can verify.
- We demonstrate a diverse set of protocols $\mathcal{P}_{\text{VPPC+DLT}}$:
 - Internet voting protocol (Applied Sciences, 2020, 100pkt) [1].
 - Privacy-Preserving, Scalable Blockchain-Based Monitoring (Appl Sci, 2022, 100pkt) [2].
 - Compliant Privacy Mixers (IACR preprint, 2025, 0pkt) [3].
 - Anonymous Provision of Privacy-Sensitive Services (IJIS, 2025, 100pkt) [4].
 - FDKG-based internet voting protocol (FGCS, 2025, 140pkt) [5].
 - Privacy Preserving Machine Learning (TASKQ, 2025, 20pkt) [6], [7].

First 5 years of my PhD: Sensitive services, Protocols, Implementations, and Articles



Sensitive services and protocol variants

Service $s \in \mathcal{S}_{sens}$		Protocol variants π (witnesses for Thesis 1)	Source
s_{ivot}	Internet Voting	p_1 centralized distributor + centralized key-holder	[pub]
		p_2 threshold distributor + threshold key-holder	[diss]
		p_3 FDKG-based voting	[pub]
s_{ppml}	PP Machine Learning	p_1 HE-BC	[diss]
		p_2 TEE-BC	[diss]
		p_3 PPML-BC*	[diss]
s_{iot}	IoT Monitoring	p_1 single key-holder + single aggregator	[pub]
		p_2 many key-holders + MPC aggregator	[diss]
s_{mixer}	Compliant Mixer	p_1 Haze	[pub]
		p_3 Haze-thresholded	[diss]
$s_{anonser}$	Anonymous Services	p_1 baseline + single court	[pub]
		p_2 baseline + DRS (e.g., Kleros)	[pub]
		p_3 Anonser-BC^{MV} (digital-verifiable + multi-provider)	[diss]

[pub] published / implemented earlier [diss] variant defined in this dissertation

Thesis 1

Combining VPPC with DLT enables building protocols for sensitive services that guarantee **CORR**, **PRIV**, and **LIVE**, without a single trusted third party.

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- Combining VPPC with DLT enables building protocols for sensitive services that guarantee **CORR**, **PRIV**, and **LIVE**, without a single trusted third party.
- Formal statement:

$$\forall s \in \mathcal{S}_{sens} \exists \pi \in \mathcal{P}_{VPPC+DLT} \forall \phi \in \{\text{CORR}, \text{PRIV}, \text{LIVE}\} : \text{TMQ}_{\phi}(\pi, s) > 1.$$

- For each $s \in \mathcal{S}_{sens}$ we provide at least one **witness protocol** π (bold variants) and a TMQ evaluation showing $\text{TMQ}_{\phi}(\pi, s) > 1$ for all three properties.

Trust Model Quantification (TMQ)

- TMQ maps protocols to a trust profile quantifying the level of decentralization of trust required to break security properties.
- TMQ says **how many parties need to collude/corrupt** to break each property.
- Introduced in “IVMF: Internet Voting Maturity Framework” (ACM DLT, 2026?, ??pkt, ?second round of minor review?) [8].
- In general, $TMQ : \mathcal{P} \rightarrow \mathbb{N}^{|\Phi|}$, where Φ is the selected set of properties.
- In this work, $TMQ : \mathcal{P}_{VPPC+DLT} \rightarrow \{L_{CORR}, L_{PRIV}, L_{LIVE}\}$.

$$TMQ : \mathcal{P}_{\mathbf{VPPC+DLT}} \rightarrow \{L_{\text{CORR}}, L_{\text{PRIV}}, L_{\text{LIVE}}\}$$

- **Correctness:** a coalition can make an honest party accept an output inconsistent with the intended logic and public transcript (without cryptographic breaks).
- **Privacy:** a coalition can learn protected inputs (or a protected link) beyond what is implied by public transcript and legitimate outputs.
- **Liveness:** a coalition can keep an honest request pending indefinitely (no output and no terminal abort), by withholding an action on the critical path.
- L_ϕ - the level of a minimal collusion set of parties that can break property ϕ .

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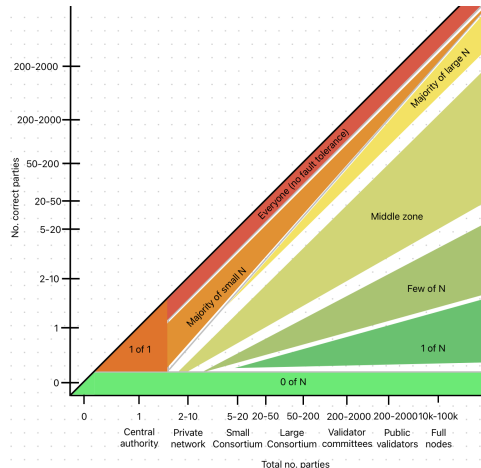
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- $L_\phi = k$ there exists a collusion of k parties that can break property ϕ , but no collusion of $k - 1$ parties can. **In protocol analysis we often do not know the exact values, so we use asymptotic categories.**
- $L_\phi = 1$ — one party can break property ϕ (centralized trust).
- $L_\phi = 2$ — a small fixed set of parties playing different roles must collude.
- $L_\phi = 3$ — a threshold of a fixed committee playing the same roles must collude.
- $L_\phi = 4$ — an open consensus majority (e.g., validators) must collude.
- $L_\phi = 5$ — a pure cryptographic failure (e.g., DLP, preimage of CRHF, factorisation, pairing).



TMQ Evaluation Procedure (Design $\rightarrow$ Capabilities $\rightarrow$ Views $\rightarrow$ Minimal Collusions $\rightarrow$ Levels) I

For each protocol π we apply the following procedure to obtain

$$\text{TMQ}(\pi) = \{L_{\text{CORR}}(\pi), L_{\text{PRIV}}(\pi), L_{\text{LIVE}}(\pi)\}.$$

The steps are written to support independent replication and inter-rater reliability.

1. Step 0: Fix background assumptions.
2. Step 1: Collect source material and resolve ambiguities..
3. Step 2: Identify threat model and define security properties (CORR/PRIV/LIVE).
4. Step 3: Identify all parties/roles $\mathcal{R}_A$.
5. Step 4: Specify the service function and the public transcript interface. Write $(y_1, \dots, y_n, T'_{\text{pub}}) \leftarrow f_A(x_1, \dots, x_n, T_{\text{pub}})$
6. Step 5: For each party $i \in \mathcal{R}_A$, write $\text{View}_i = (x_i, r_i, m_i^{\text{in/out}}, T_{\text{pub}})$.

TMQ Evaluation Procedure (Design $\rightarrow$ Capabilities $\rightarrow$ Views $\rightarrow$ Minimal Collusions $\rightarrow$ Levels) II

7. Step 6: Define the goal formulas $G_X(A)$ for $X \in \{\text{CORR}, \text{PRIV}, \text{LIVE}\}$.

Write (per property) a short operational scope paragraph, then give a Boolean goal formula over primitives. Goals must be stated in terms of *Read/Write/Withhold/Rewrite* only.

8. Step 7: Compute minimal collusion sets by satisfying goals.

For each property X :

- enumerate candidate collusion sets S and prove $(\bigcup_{i \in S} \text{Cap}(i)) \models G_X(A)$;
- inclusion-minimise these sets to obtain $\mathcal{M}_X(A)$.

9. Step 8: Map each minimal set to a lattice level and take the weakest.

For each $S \in \mathcal{M}_X(A)$ map its structure to $\ell(S) \in \mathcal{L}$. $L_X(A) = \min_{S \in \mathcal{M}_X(A)} \ell(S)$.

10. Step 9: Evaluate centralized counterpart (when applicable).

11. Step 10: Multi-rater application (IRR).

What we show for Thesis 1

- **Assumed (VPPC building blocks):**
 - VPPC mechanisms used by a variant satisfy their standard security definitions (e.g., soundness/zero-knowledge, privacy/correctness of MPC/threshold crypto, correctness of HE, validity of TEE attestation).
- **Shown in this dissertation (TMQ):**
 - For each protocol variant π we define roles, views, and goal formulas for CORR/PRIV/LIVE.
 - We prove **minimal collusion sets** of trust domains that can break each property, and map them to levels $L_{\text{CORR}}(\pi)$, $L_{\text{PRIV}}(\pi)$, $L_{\text{LIVE}}(\pi)$.
- **For variants marked [diss] in the table:** we additionally provide a precise protocol description and explicit assumptions, then apply the same TMQ procedure (capability level).

Results for Thesis 1

$$\forall s \in \mathcal{S}_{sens} \exists \pi \in \mathcal{P}_{VPPC+DLT} \forall \phi \in \{\text{corr, priv, live}\} : \text{TMQ}_{\phi}(\pi, s) > 1$$

Service / Variant	CORR	PRIV	ROB	PATTERNS
Centralized Stellot	1	1	1	BLINDSIG, NULL, PKE, MERKLE, REG
Stellot ⁺	4	1	1	POE, BLINDSIG, NULL, PKE, MERKLE, REG
Stellot [*]	4	3	1	POE, BLINDSIG, NULL, PKE, MERKLE, REG, CPV, TC, LHE, RANGE
Stellot [†]	4	3	3	POE, BLINDSIG, NULL, PKE, MERKLE, REG, CPV, TC, LHE, RANGE
FDKG-Vote-BC	4	3	3	CPV, TC, LHE, RANGE
Centralized PPML-BC-VPPC	1	1	1	–
PPML-TEE-BC-TEE	1	2	1	POE, CPV, PKE
PPML-HE-BC-HE	1	5	1	POE, PKE
PPML-BC*-BC-HE-zkML	4	5	2	POE, CPV, PKE
Centralized mixer	1	1	1	–
CompliantMixer-BC _{cen} Daze	4	1	1	CPV, MERKLE, NULL, REG, PKE
CompliantMixer-BC _{cen} Haze	4	5	1	CPV, MERKLE, NULL, REG
CompliantMixer-BC _{thr} Daze	4	3	3	CPV, MERKLE, NULL, REG, PKE, TC
CompliantMixer-BC _{thr} Haze	4	5	3	CPV, MERKLE, NULL, REG, TC
Centralized Anonser	1	5	1	PKE, DISPUTE, EVID
Anonser-BC	1	5	1	POE, PKE, DISPUTE, EVID
Anonser-BC ^M	3	5	2	POE, PKE, DISPUTE, EVID
Anonser-BC ^{VPPC}	4	5	1	POE, CPV, PKE, DISPUTE, EVID
Anonser-BC ^{MV}	4	5	2	POE, CPV, PKE, DISPUTE, EVID

Thesis 2: Problem of DKG in large/open networks

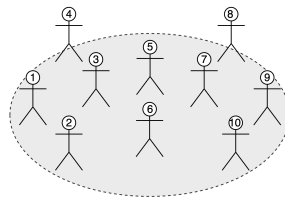
- Many $\mathcal{P}_{\text{VPPC+DLT}}$ need a key operation on the critical path: *decrypt tally, sign authorization, release secret, rotate committee key*.
- Replacing a single key holder with Distributed Key Generation (DKG) moves the minimal break from *one party* to a *threshold of parties*.

$$L_{\text{PRIV}}(\pi_{\text{DKG}}, s) = 3 \quad \text{whereas} \quad L_{\text{PRIV}}(\pi_{\text{single}}, s) = 1.$$

- However, classical (t, n) -DKG is not scalable in open / large networks because it assumes a fixed set and high timely participation; when participation drops, liveness degrades or setup aborts.
- "Getting 60% of 10 people online is easy; getting 60% of 1,000 people online in the same window is hard"

Thesis 2

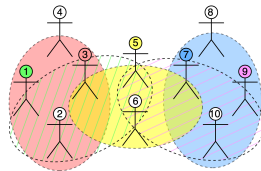
- Published in Federated Distributed Key Generation (FGCS, 2025, 140pkt) [5].
- FDKG replaces one global quorum with many local quorums (guardian sets), enabling large- n deployments with optional participation, while keeping one-round PVSS-style broadcasts.
- Thesis 2:** Using DKG strengthens PRIV and LIVE for $\mathcal{P}_{VPPC+DLT}$ by removing single key holders. Federated DKG (FDKG) generalizes DKG to optional participation and heterogeneous trust, improving practical robustness for large n , while preserving one-round PVSS-style communication.



$$\text{DKG is a set } \mathbb{D} = \left\{ \begin{array}{cccc} 1 & 2 & 3 & 4 & 5 \\ 6 & 7 & 8 & 9 & 10 \end{array} \right\}$$

$$\text{Node } i \text{'s Guardian Set is } \mathbb{D} \setminus \{i\}$$

$$\text{Liveness achieved by } \{L \subseteq \mathbb{D} \mid |L| \geq 5\}$$



$$\text{FDKG is a set } \mathbb{D} = \left\{ \begin{array}{cccc} 1 & 2 & 3 & 4 & 5 \\ 6 & 7 & 8 & 9 & 10 \end{array} \right\}$$

$$\text{Node } 1 \text{'s Guardian Set } G_1 \text{ is } \{2, 3, 4, 5\}$$

$$\text{Node } 3 \text{'s Guardian Set } G_3 \text{ is } \{1, 2, 4, 5\}$$

$$\text{Node } 5 \text{'s Guardian Set } G_5 \text{ is } \{3, 4, 6, 7\}$$

$$\text{Node } 7 \text{'s Guardian Set } G_7 \text{ is } \{4, 5, 8, 9\}$$

$$\text{Node } 9 \text{'s Guardian Set } G_9 \text{ is } \{5, 6, 7, 10\}$$

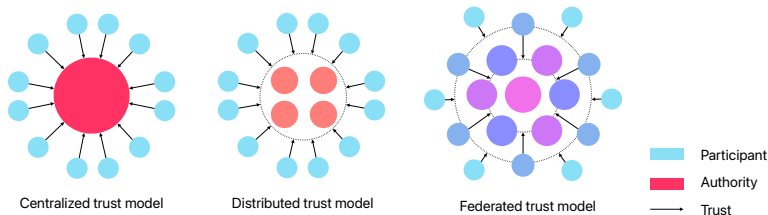
$$\text{Liveness achieved by } \{\{3, 5, 7\}, \{1, 3, 5, 7\}, \{3, 5, 7, 9\}, \{1, 3, 5, 7, 9\}\}$$

FDKG: replace one global quorum with many local quorums

- Classical DKG: **one global requirement** (enough of the whole committee online at once).
- FDKG: **local recoverability** for each contributor: each participant i chooses a guardian set G_i (size k) and a local threshold t .
- Reconstruction succeeds if each contribution is recoverable:

$$\forall i \in D : (i \text{ reveals}) \vee (|T \cap G_i| \geq t).$$

- The coordination burden is controlled by k (local), not by n (global).



What we show for Thesis 2

- **DKG:** Replacing a single key holder with DKG-based threshold cryptography increases L_{PRIV} and often L_{LIVE} .
- **Security:** We provide formal proofs FDKG is secure (correctness, privacy, and liveness) under standard assumptions.
- **Scalability:** We show by simulation that FDKG can achieve higher success probability than classical DKG under probabilistic participation.
- **Cost:** FDKG keeps one-round broadcast in the setup phase; communication scales as $O(|D| \cdot k)$.
- **Application:** We integrate FDKG with internet voting protocol.

Results for Thesis 2

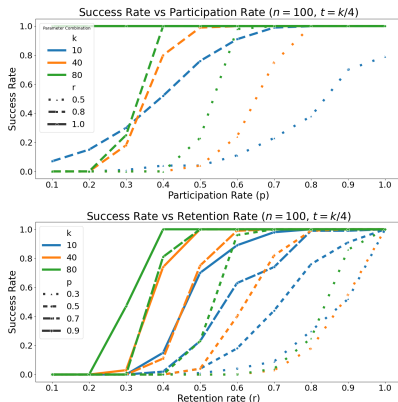


Figure: Success vs. participation p (top) and retention r (bottom) at $n = 100$ with $t = k/4$. Line styles encode the

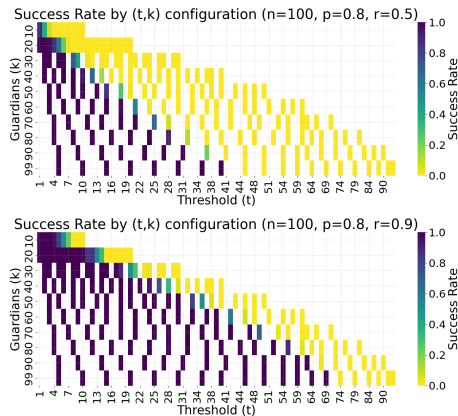


Figure: Success over (k, t) for $n = 100$, $p = 0.8$, with $r = 0.5$ (top) and $r = 0.9$

Main contributions

- **A set of sensitive services and protocol variants.**
 - Defined $\mathcal{S}_{sens} = \{s_{ivot}, s_{ppml}, s_{iot}, s_{anonser}, s_{mixer}\}$ and, for each s , designed a set of DLT+VPPC protocol variants with explicit roles, transcripts, and assumptions.
 - Identified **witness variants** (bold in the table) that achieve $TMQ_{\phi}(\pi, s) > 1$ for $\phi \in \{\text{CORR}, \text{PRIV}, \text{LIVE}\}$.
- **Trust Model Quantification (TMQ): a replicable method to compare trust models.**
 - Introduced a step-by-step evaluation pipeline: Design $\rightarrow$ Views $\rightarrow$ Capabilities $\rightarrow$ Goal formulas $\rightarrow$ Minimal collusion sets $\rightarrow$ Levels.
 - Produced a **TMQ coding sheet** for each variant and a summary table $TMQ(\pi) = \{L_{\text{CORR}}, L_{\text{PRIV}}, L_{\text{LIVE}}\}$ enabling cross-protocol comparison.
- **New cryptographic primitive for open/large networks: Federated DKG (FDKG).**
 - Designed and analyzed FDKG: replaces one global quorum with many local guardian quorums, enabling optional participation and heterogeneous trust.
 - Demonstrated that replacing a single key holder with (F)DKG strengthens **PRIV** and often **LIVE** by raising the minimal collusion level.
- **Practical evidence: protocols and implementations.**

Limitations

- We don't provide a unified architecture that achieves all properties for all sensitive services.
- The security properties are evaluated at a high (capability) level.
- Some level boundaries (1..5) can be blurry in edge cases (e.g., censoring vs rewriting, “few/1” vs “closed t/n ”).
- I only need the claim $\mathbf{TMQ}_\phi > 1$ for the thesis, so ultra-sharp borders beyond that are not essential.

Status & next steps (roadmap to finish the dissertation)

- **Step 1: Freeze scope and assumptions.**
 - lock $\mathcal{S}_{sens}$, the evaluated protocol variants, and the operational scope of CORR/PRIV/LIVE;
 - lock background assumptions (network model, “no cryptographic breaks”, meaning of terminal abort).
- **Step 2: Finish TMQ coding for each protocol variant.**
 - for each variant: define **Views** $\rightarrow$ derive **Capabilities** $\rightarrow$ state **goal formulas** $\rightarrow$ prove **minimal collusion sets** $\rightarrow$ map to levels $L_{CORR}, L_{PRIV}, L_{LIVE}$.
- **Step 3: Finish the property-by-property arguments.**
 - for each variant: short CORR/PRIV/LIVE argument aligned with the TMQ abstraction (capability level).
- **Step 4: Fill the TMQ results table with coding-sheet-backed entries.**
 - every table cell must be traceable to a concrete minimal-collusion argument in the coding sheet.
- **Suggested order (fast momentum):** $s_{ivot}(p_2) \rightarrow s_{ivot}(p_3) \rightarrow s_{iot}(p_2) \rightarrow s_{mixer}(p_3 \text{ or } p_4) \rightarrow s_{anonser}(p_3) \rightarrow s_{ppml}(p_3)$.

Future work

- Applications to other fields: cybersecurity, forensics, criminal justice, law, anywhere trust and corruption are at stake.
- A double-edged sword: the methodology provides insights for both defenders and attackers.

Dziękuję!

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